Write your name here

Surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9-1)

Centre Number	Candidate Number

Chemistry

Paper 1

Foundation Tier

Sample Assessment Materials for first teaching September 2016
Time: 1 hour 45 minutes

Paper Reference
1CH0/1F

You must have:
Calculator, ruler

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – *there may be more space than you need.*
- Calculators may be used.
- Any diagrams may NOT be accurately drawn, unless otherwise indicated.
- You must **show all your working out** with **your answer clearly identified** at the **end of your solution**.

Information

- The total mark for this paper is 100
- The marks for **each** question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- In questions marked with an asterisk (*), marks will be awarded for your ability to structure your answer logically showing how the points that you make are related or follow on from each other where appropriate.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

S50045A

©2016 Pearson Education Ltd.

S 5 0 0 4 5 A 0 1 3 2

PEARSON

Answer ALL questions. Write your answers in the spaces provided.

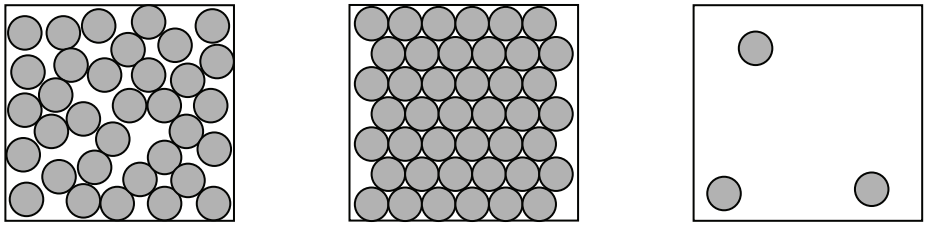
Some questions must be answered with a cross in a box ☒.
If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 There are three states of matter, solid, liquid and gas.

(a) The three boxes in Figure 1 show the arrangement of particles in different states.

(i) Under each box write the name of the state of matter shown.

(2)



....

Figure 1

(ii) A student is given some solid wax.

Use words from the box to name **two** pieces of equipment that the student should use to convert the solid wax into a liquid.

Bunsen burner	test tube	filter funnel
burette	pipette	

(2)

1 .....

2 .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 ©
Pearson Education Limited 2016

(b) Some liquid is left in a warm room.

After a few days no liquid can be seen.

Give the name of the process that has occurred.

(1)

.....

(c) The freezing point of water is 0 °C.

(i) Describe how the movement and arrangement of water particles changes when water is cooled from 10 °C to –10 °C.

(2)

.....

.....

.....

.....

(ii) What is the structure of water?

(1)

☒ **A** ionic

☒ **B** simple molecular (covalent)

☒ **C** giant covalent

☒ **D** metallic

(Total for Question 1 = 8 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2 Ammonium phosphate and ammonium sulfate are made from ammonia.

These compounds can be used as fertilisers.

(a) Ammonia solution is alkaline.

Which of the following could be used to show that ammonia solution is alkaline?

(1)

☒ **A** conical flask

☒ **B** pH meter

☒ **C** pipette

☒ **D** thermometer

(b) Give **one** advantage of using fertilisers made from ammonia rather than using manure.

(1)

.....

.....

(c) The fertiliser ammonium phosphate was made by reacting ammonia solution with dilute phosphoric acid.

(i) In the first step, 25 cm^3 of dilute phosphoric acid was placed in a beaker.

Give the name of a piece of apparatus that could be used to measure out the 25 cm^3 dilute phosphoric acid.

(1)

.....

(ii) Complete the word equation for this reaction.

(1)

ammonia + ... → .....

(iii) Some ammonium phosphate solution was made.

Describe how pure, dry crystals of ammonium phosphate are obtained from the ammonium phosphate solution.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

(d) The formula of ammonium sulfate is $(NH_4)_2SO_4$.

What is the empirical formula of ammonium sulfate?

(1)

☒ **A** NHSO

☒ **B** NH_2SO_2

☒ **C** NH_4SO_4

☒ **D** $N_2H_8SO_4$

(Total for Question 2 = 7 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3 Unreactive metals are found as uncombined metals in the Earth's crust.

(a) Which of the following metals is found uncombined in the Earth's crust?

(1)

☒ **A** aluminium

☒ **B** gold

☒ **C** sodium

☒ **D** zinc

(b) When iron oxide is heated with carbon, iron is produced.

(i) Complete the word equation for the reaction.

(2)

iron oxide + carbon → ..... + ...

(ii) What happens to the iron oxide during this reaction?

(1)

☒ **A** the iron oxide burns

☒ **B** the iron oxide is neutralised

☒ **C** the iron oxide is oxidised

☒ **D** the iron oxide is reduced

(c) Copper ore contains copper carbonate, $CuCO_3$.

In the first stage of the extraction process, the copper carbonate is decomposed by heating to form copper oxide, CuO, and carbon dioxide.

$$CuCO_3 \rightarrow CuO + CO_2$$

When 100 g of copper carbonate is decomposed completely in this way, it is found that the total mass of products is 100 g.

Give a reason why the starting mass of copper carbonate is **always** the same as the mass of the products formed.

(1)

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(d) Zinc can be extracted from its ore by electrolysis or by heating the ore with carbon.

Give a reason for the method that is used.

(1)

.....

.....

(e) Figure 2 gives information about aluminium and tin.

metal	cost of 1 kg / £	amount in Earth's crust / %
aluminium	1.31	8
tin	12.60	0.0002

Figure 2

Give **two** reasons why it could be more important to recycle tin than to recycle aluminium. Use the information in Figure 2.

(2)

Reason 1

.....

.....

Reason 2

.....

.....

(Total for Question 3 = 8 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

BLANK PAGE

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4 An electrolysis experiment is carried out on different solutions, **J**, **K** and **L**.

Electricity is passed through each solution as shown in Figure 3.

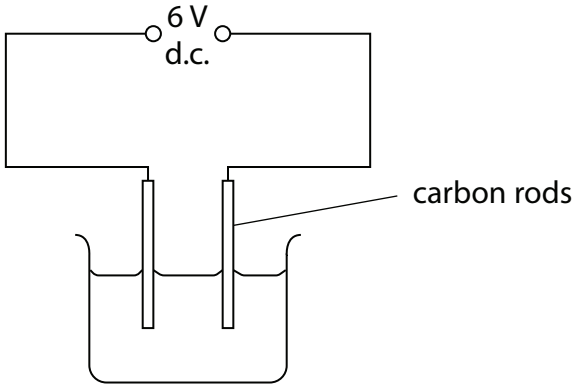


Figure 3

Any products formed at the electrodes are identified.

The results are given in Figure 4.

solution	solution conducts electricity	product at cathode	product at anode
J	yes	copper	chlorine
K	no	none	none
L	yes	hydrogen	chlorine

Figure 4

(a) (i) State an improvement that can be made to the circuit to show that a current is flowing during the electrolysis.

(1)

.....

Some of these solutions are electrolytes.

(ii) State what is meant by the term **electrolyte**.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(iii) Which of **J**, **K** and **L** are electrolytes?

(1)

☒ **A** **K** only

☒ **B** **J** and **L** only

☒ **C** **K** and **L** only

☒ **D** **J**, **K** and **L**

(b) Copper sulfate solution was electrolysed for five minutes using copper electrodes.

Figure 5 shows the mass of the anode and of the cathode before electrolysis and after electrolysis.

	anode	**cathode**
mass of electrode before electrolysis / g	1.16	1.28
mass of electrode after electrolysis / g	0.85	1.57

Figure 5

Calculate the mass of copper deposited.

(2)

mass of copper deposited = ... g

(c) Identify the products formed at the anode and cathode when molten potassium iodide is electrolysed.

(2)

Anode .....

Cathode .....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(d) In a different electrolysis, molten sodium fluoride is decomposed.

$$2NaF \rightarrow 2Na + F_2$$

(relative atomic masses: F = 19, Na = 23)
(relative formula mass NaF = 42)

Calculate the maximum mass of sodium that could be formed from 168 g of sodium fluoride.

(2)

mass = ... g

(Total for Question 4 = 10 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5 (a) Give **two** advantages for electroplating some metal objects.

(2)

.....

.....

(b) Solder is an alloy of tin and lead.

A sample of a solder was made by mixing 22.5 g of lead with 15.0 g of tin.

Calculate the percentage of tin in this solder.

(2)

percentage of tin = ... %

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(c) The rusting of an iron nail was investigated by setting up three test tubes, as shown in Figure 6.

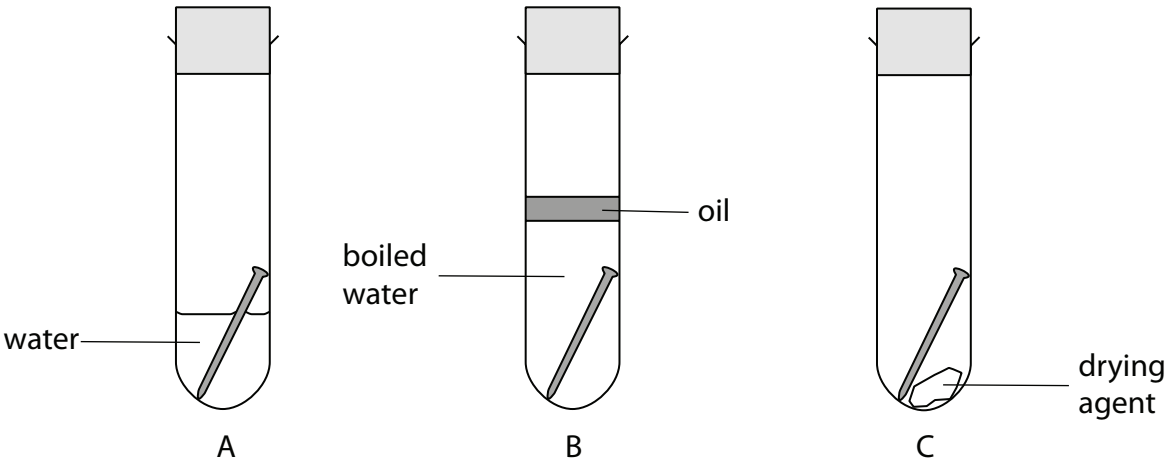


Figure 6

State and justify the result you would see in each tube after one week.

(3)

A .....

.....

B .....

.....

C .....

.....

(d) An iron bucket is coated in zinc.
Over many years of use, the iron bucket has been scratched and left outside in the rain. Although some of the zinc coating has been removed to expose iron, the iron bucket has not rusted.

Explain why the iron has not rusted.

(2)

.....

.....

.....

.....

(Total for Question 5 = 9 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6 The apparatus in Figure 7 shows a piece of magnesium ribbon being heated.

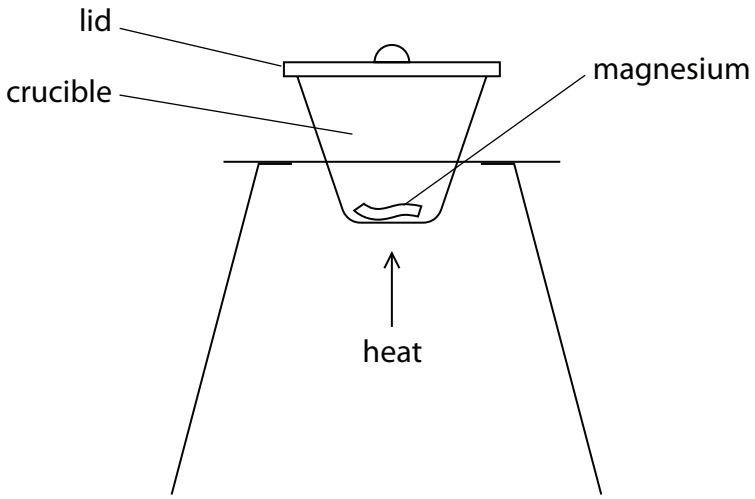


Figure 7

During the heating, the magnesium reacts with oxygen from the air. The lid of the crucible was raised slightly from time to time. Magnesium oxide was formed as a white powder. The experiment was repeated with different masses of magnesium.

The results are shown in Figure 8.

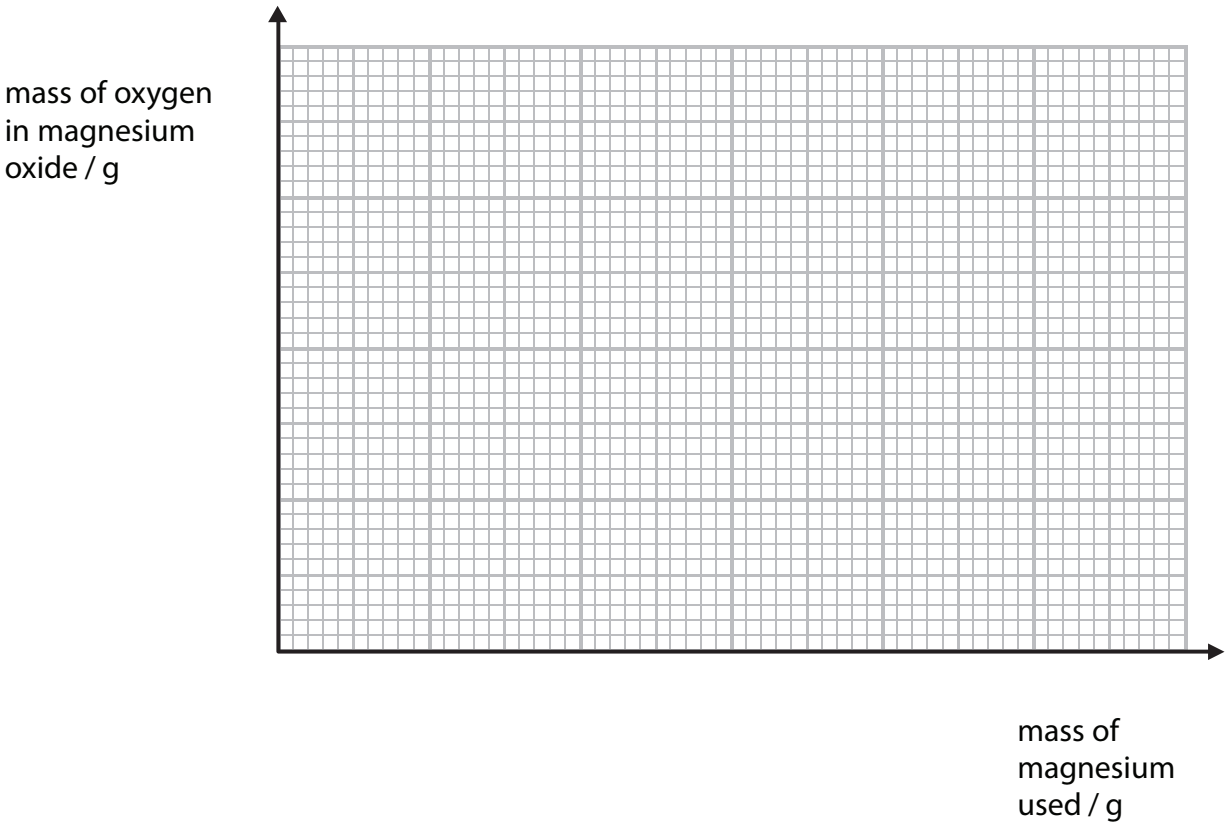
experiment	mass of magnesium used / g	mass of magnesium oxide formed / g	mass of oxygen in magnesium oxide / g
1	0.10	0.16	0.06
2	0.15	0.24	0.09
3	0.25	0.40	0.15
4	0.30	0.48	0.18
5	0.35	0.49	0.14
6	0.50	0.80	0.30

Figure 8

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(a) (i) Draw a graph of the mass of oxygen in magnesium oxide against the mass of magnesium used.

(3)



(ii) The result for experiment 5 is anomalous.
The masses were all measured accurately.

Suggest a reason for this anomalous result.

(1)

.....

.....

(b) Balance the equation for the reaction of magnesium with oxygen to form magnesium oxide.

(1)

..... Mg + ... O_2 → ..... MgO

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(c) Calcium nitrate contains calcium ions and nitrate ions.

Calculate the relative formula mass of calcium nitrate, $Ca(NO_3)_2$.
(relative atomic masses: Ca = 40, N = 14, O = 16)

(2)

relative formula mass = ...

(d) Two oxides of lead, **R** and **S**, were analysed.

The empirical formula of oxide **R** was found to be PbO.

The results of the analysis of oxide **S** showed it contained 0.207 g of lead combined with 0.032 g of oxygen.

Show, by calculation, that the two oxides had different empirical formulae.
(relative atomic masses: O = 16, Pb = 207)

(3)

(Total for Question 6 = 10 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 ©
Pearson Education Limited 2016

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

BLANK PAGE

7 Substances can be pure or they can be mixtures.

(a) Which of these is a mixture?

(1)

☒ **A** chlorine

☒ **B** sodium

☒ **C** sodium chloride

☒ **D** sodium chloride solution

(b) Figure 9 shows some mixtures to be separated and possible methods of separation.

Place a tick (✓) in one box in each row of the table to show the best method to separate the first named substance from each of the mixtures.

(3)

substance to separate	method of separation			
	crystallisation	filtration	simple distillation	fractional distillation
sand from a mixture of sand and sodium chloride solution				
copper sulfate crystals from copper sulfate solution				
useful liquids from crude oil				

Figure 9

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

(c) Paper chromatography was used to separate a mixture of blue and red inks. A spot of the mixture was placed on chromatography paper as shown in Figure 10.

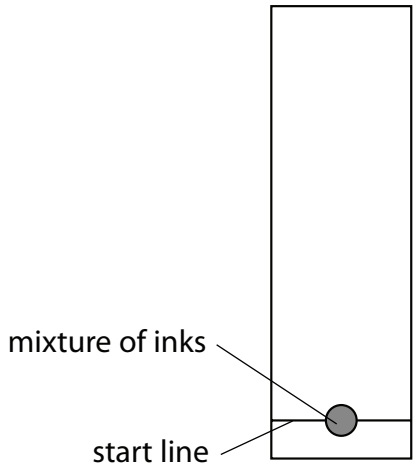


Figure 10

(i) Give a reason why the start line is drawn in pencil rather than in ink.

(1)

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(ii) The chromatography paper, with the spot of mixture on it, was placed in a beaker with the bottom of the paper in solvent.

On Figure 11, complete the diagram showing the position of the chromatography paper with the spot of mixture at the start of the experiment.

(1)

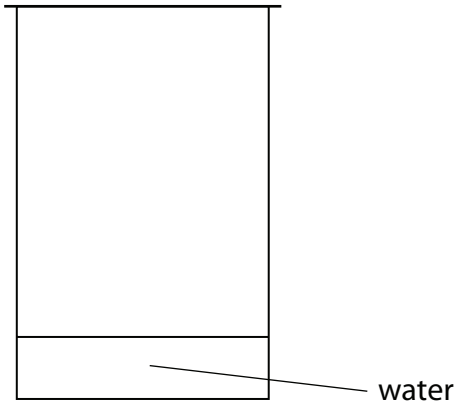


Figure 11

(iii) The chromatography was carried out and the result is shown in Figure 12.

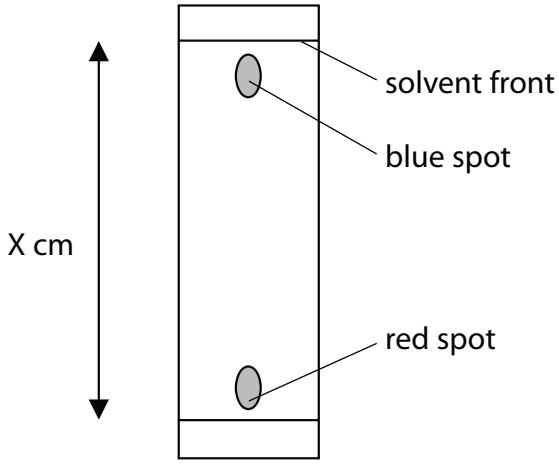


Figure 12

The blue spot had moved 14.5 cm and the solvent front had moved 15.3 cm.

Calculate the R_f value of the substance in the blue spot, giving your answer to 2 significant figures.

$$R_f \text{ value} = \frac{\text{distance travelled by a dye}}{\text{distance travelled by solvent front}}$$

(2)

R_f value = ...

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

(d) P, Q, R and S are mixtures of food colourings.

They are investigated using paper chromatography.

Figure 13 shows the chromatogram at the end of the experiment.

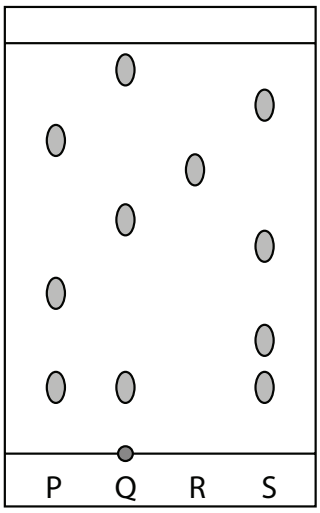


Figure 13

(i) Which mixture contains an insoluble food colouring?

(1)

☒ **A** mixture **P**

☒ **B** mixture **Q**

☒ **C** mixture **R**

☒ **D** mixture **S**

(ii) Give a change that could be made to the experiment to obtain an R_f value for the insoluble colouring.

(1)

.....

.....

(iii) Explain, by referring to Figure 13, which mixture is separated into the greatest number of soluble food colourings by this chromatography experiment.

(2)

.....

.....

.....

.....

(Total for Question 7 = 12 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8 A titration is to be carried out to find the concentration of a solution of sodium hydroxide.

The sodium hydroxide solution is titrated with dilute sulfuric acid.

The available apparatus includes a burette, a pipette, a funnel, a conical flask and an indicator.

(a) State one safety precaution that must be taken when using sodium hydroxide solution and dilute sulfuric acid.

(1)

.....

.....

.....

(b) The sodium hydroxide solution is made by dissolving 4.3 g of sodium hydroxide in water and making the solution up to 250 cm^3 with water.

Calculate the concentration of the solution in g dm^{-3}.

(2)

concentration = ... g dm^{-3}

(c) Write the balanced equation for the reaction of dilute sulfuric acid, H_2SO_4, with sodium hydroxide.

(2)

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

(d) The results of titrations to determine how much of an acid is required to neutralise a given volume of an alkaline solution are shown in Figure 14.

	titration 1	titration 2	titration 3	titration 4
final burette reading (cm^3)	27	27.40	29.20	29.30
initial burette reading (cm^3)	0	2.10	4.00	3.50
volume of acid used (cm^3)	27	25.30	25.20	25.80

Figure 14

Two of the titrations in Figure 14 should **not** be used to calculate the mean volume of acid required.

Identify each titration and give a reason why it should not be used in the calculation of the mean.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

*(e) Describe the experimental procedure to carry out a titration to find the exact volume of sulfuric acid needed to neutralise 25.0 cm^3 of sodium hydroxide solution and obtain pure, dry crystals of sodium sulfate.

(6)

(Total for Question 8 = 13 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

9 Ionic compounds contain ions.

(a) The numbers of electrons, neutrons and protons in four particles, **W**, **X**, **Y** and **Z**, are shown in Figure 15.

particle	electrons	neutrons	protons
W	9	10	9
X	10	14	12
Y	16	16	16
Z	18	18	16

Figure 15

Explain which particle, **W**, **X**, **Y** or **Z**, is a negative ion.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(b) Lithium fluoride, LiF, is an ionic compound.

It contains lithium cations and fluoride anions.

The electronic configurations of a lithium atom and of a fluorine atom are shown in Figure 16.

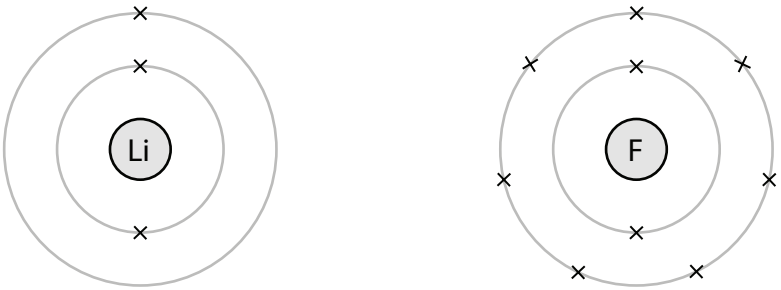


Figure 16

Complete Figure 17 to show the electronic configurations and charges of the ions in lithium fluoride.

(4)



charge on ion ..... charge on ion .....

Figure 17

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

*(c) Figure 18 shows the ability of different substances to conduct electricity.

substance	conducts electricity
solid calcium chloride	no
molten calcium chloride	yes
diamond	no
zinc	yes

Figure 18

Explain these results by referring to the structures of the substances.

(6)

(Total for Question 9 = 12 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10 The method used to prepare a salt depends on its solubility in water.

(a) Complete Figure 19 by placing one tick in each row to show whether the salt is soluble or insoluble.

(2)

salt	soluble	insoluble
ammonium chloride		
lithium sulfate		
magnesium carbonate		

Figure 19

(b) Lead nitrate solution mixed with sodium sulfate solution forms lead sulfate as a precipitate.

$$Pb(NO_3)_2 + Na_2SO_4 \rightarrow PbSO_4 + 2NaNO_3$$

In an experiment, the theoretical yield of lead sulfate for this reaction was 2.85 g. In the experiment only 2.53 g of lead sulfate is obtained.

Calculate the percentage yield of lead sulfate in this experiment.

Give your answer to two significant figures.

(3)

percentage yield = ...%

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016

(c) The method used to make the lead sulfate is:

- pour 100 cm^3 lead nitrate solution into a beaker
- add drops of sodium sulfate solution until a precipitate is seen
- allow the precipitate to settle to the bottom of the beaker
- pour off the liquid
- use a spatula to transfer the solid lead sulfate onto a filter paper.

Explain **two** ways of improving this experimental method in order to increase the amount and quality of lead sulfate that could be obtained from 100 cm^3 of lead nitrate solution.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(d) Ammonium nitrate is produced from ammonia and nitric acid on a large scale in industry.

Ammonium nitrate can also be made in the laboratory by titrating ammonia solution with dilute nitric acid.

$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$

Ammonium nitrate crystals can then be obtained by evaporating off some of the water from the solution.

Give **two** reasons why this laboratory method is not suitable for use on a large scale in industry.

(2)

.....

.....

.....

.....

(Total for Question 10 = 11 marks)

TOTAL FOR PAPER = 100 MARKS

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Chemistry – Sample Assessment Materials – Issue 1 – March 2016 © Pearson Education Limited 2016